

# Sacred water: rediscovering the ancient hydraulic system of Angkor and traditional knowledge of water management and engineering systems

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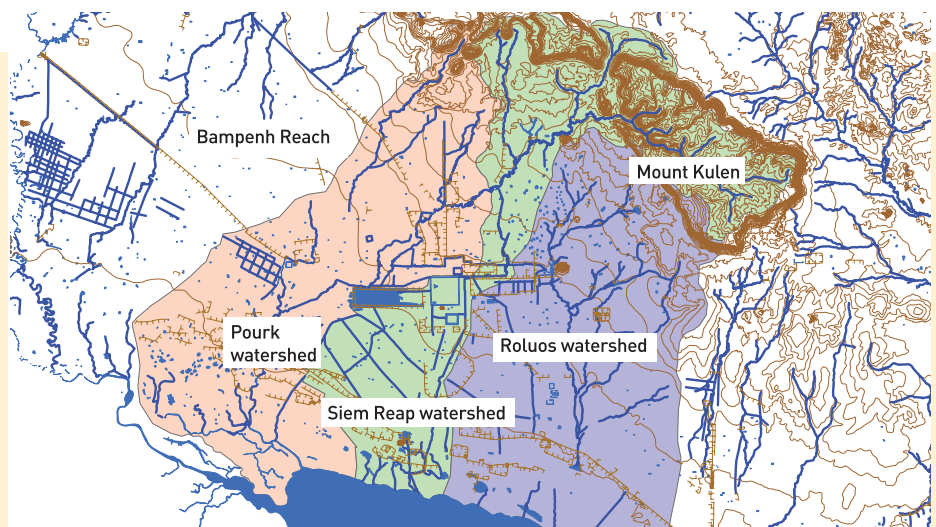
## 1.Introduction

The Angkor region was the capital of the Khmer Empire for more than 500 years. Angkor was inscribed on the World Heritage List in 1992; it is a living site with 112 villages. (Plates 1 and 2) Angkor not only has its temples but also intangible heritage that is being discovered every day by researchers.

Angkor was recognised by Bernard-Phillip Groslier (of the École Française d'Extrême-Orient – EFEO) in the 1950s and 1960s as the 'Hydraulic City' because the city is



**Plates 1-2**  
Angkor Wat, aerial and general views. Photos: Author



**Figure 1**  
Watersheds of the Roluos (East), Siem Reap (Centre) and Pourk (West) rivers. Source: Author

organised around an immense water management network (Groslier; 1979), but the functioning of this water network was not understood at that time. It was only after 2004-2005 that Khmer researchers discovered how it functions.

The Khmer people have lived with water since the creation of Cambodia in the 1<sup>st</sup> century. Water management is a part of their daily life, and the evidence in the Angkor region shows that the innovative water technology they were using was cutting edge. Water management was considered to be basic knowledge that everyone understood which is why nothing was written about it. The best way to find out about it, more than 500 years later, is to discover what local people remember and what memories have been transmitted down the generations.

This article will bring together two ways of understanding the ancient Khmer hydraulic system - by combining Khmer cultural knowledge of traditional water management and modern knowledge (with hydraulic and hydrological modelling) to rehabilitate this ancient system to safeguard the temples in the Angkor World Heritage property.

## 2. Water in Khmer culture

The Siem Reap/Angkor region has two natural watersheds in the Roluos and Pourk rivers. In the 9<sup>th</sup> century, when the development at Roluos became the capital, the demand for water increased; to meet this

demand for water, the Siem Reap channel (the Siem Reap River) was created to divert the water from the Pourk River by constructing a laterite spillway called the Bampenh Reach. Nowadays the Angkor World Heritage site is supplied by three watersheds, the Roluos, Siem Reap and Pourk with their catchment areas of 1031.84 square kilometres, 836.74 square kilometres and 935.62 square kilometres respectively. (Figure 1) So the Siem Reap River, with its source on Mount Kulen, plays a very important role in the Angkor plain region.

### 2.1. Temple construction

The soil in the central plain of Cambodia cannot on its own support heavy buildings. In order to build stone temples like Angkor Wat, Bayon, Ta Prohm and Preah Khan, the best technique had to be found. Khmer engineers at the time discovered the physical properties of sand and water and realised that they could combine these two elements for building because sand, once wet, can support a heavy load. The discovery of this technique led them to locate the places where this theory could be applied. Studies have shown that the Angkor region is the best location as underground water is close to the surface. They then used the immediate presence of underground water to completely fill the sand layer under the monument to ensure its stability. To assure the sustainability of ground water to support the temples, the Khmer ancestors introduced the concept or idea that water was sacred and that belief is still held today.

The ancient Khmers knew the vital role played by water resources in the safeguarding of the Angkor region

and learned how to preserve water. This is why this vital resource is celebrated within the tradition, culture and spirit of the Khmer people. Some of these customs are still celebrated today.

## 2.2. The sacred water of Mount Kulen

The Khmer ancestors carved the Siem Reap River of a Thousand Lingas in the river beds of Mount Kulen and Kbal Spean where these rivers source before they flow into Siem Reap and the Angkor plain. At Banteay Srei they flow together to form the Siem Reap River. The water flowing from the 'Thousand Lingas' has become sacred and has been used in all the major ceremonies (e.g. coronations) of the Khmer Kingdom since the 9<sup>th</sup> century. During a coronation, the sacred water of Mount Kulen is used to bless the future King. This tradition is still practised. The Khmer population believes in the power of this sacred water, using it to cure diseases or in blessings to bring luck. But the real goal of designating the water from Mount Kulen as sacred was to underline to the population the need to protect water resources, the region's life-blood, and to maintain the sustainability of this resource which is essential for the conservation and development of the Siem Reap region. However, the water source of Mount Kulen will be lost if deforestation continues and the environment is destroyed.

## 2.3. The moats

Before the construction of the temples the natural soil was removed and replaced with sand to which water would be added to make it load bearing. Of course, this sand layer is linked to the groundwater, but to make sure that the temple would still have a sound foundation if there was any variance in underground water level, the moat system was adopted. Thus each temple is surrounded by a moat which plays a pivotal role in its stability: it collects the run-off water from the temple during the monsoon and recharges the sand layer underneath the temple.

The Khmers' ancestors understood that if the safeguarding of water was enforced by law and the need was explained in purely technological terms, the people would probably not listen, but by presenting the water as a form of life-blood and as the basis for a system of beliefs, people were more likely to look after the water system. So, to ensure there was water in the moat to

support the temple, what was actually an engineering requirement was transformed into a religious duty. In the Khmer tradition, the moats are considered as the Ocean and the temple as Mount Meru (the dwelling of the gods).

## 3. The water management network

Water remains a major challenge for urban planners in the big cities; the ancient Khmer had mechanisms for collecting water in the rainy season centuries ago. This can be seen in Angkor, Banteay Chhmar, Preah Vihear and on other sites. They built reservoirs, moats, ponds and retaining reservoirs. To maintain a constant supply and to store water for the town's everyday use over the six months of the dry season was a difficult task, almost impossible. Hence the digging of man-made canals to channel water, the construction of spillways, for example in Bampenh Reach, to redirect the natural flow of the River Pourk, whose source was on Mount Kulen, towards the man-made Siem Reap River in order to supply Roluos, the capital, with water.

This overflow weir (Bampenh Reach) played a dual role: first it supplied water to the city through the Siem Reap River and second it prevented any prospective flooding by sending water to the Pourk and Siem Reap rivers to prevent the inundation of Roluos. During the dry season when there is less water from Mount Kulen, the water was directed only to Siem Reap River, but in the rainy season, when there is too much water, it flows to both the River Pourk and Siem Reap River. This is evidence that Khmer engineering in the 9<sup>th</sup> century already used an automated system of water management. The excess water was stored in basins (*srang*), moats and *barays* (reservoirs).

All *barays* are used to recharge the groundwater by direct infiltration, but some *barays* have other functions too, for example, the Lolei Baray and West Baray are used for irrigation, and the Jayatataka or North Baray is used to supply Angkor Thom city. The East Baray in the first phase is used for irrigation purposes like Lolei and West Baray, the second phase is limited to the fundamental role of recharging the ground water to ensure the sustainable supply of water to the Srah Srang (Royal Basin) and moats of Ta Prom, Banteay Kdei and other temples downstream.



### 3.1. Hydraulic engineering

Several elements provide evidence that the ancient Khmers understood the way the flow of water operates. The Bampenh Reach spillway (built in the 9<sup>th</sup> century) is a good example of a typical construction; the spillway was built following the curve of the waterflow in order to reduce turbulence and erosion downstream. The ancient Khmers also understood hydraulic force, which explains the presence of the laterite blocks used to build the spillway. To prevent any movement of the blocks and to keep them in their positions despite the torrential force of water, they cut vertical or horizontal grooves into the blocks so that they would interlock with each other and form huge solid sections. (Plates 3a and b)

### 3.2. The Jayatataka (North Baray)

The Jayatataka (North Baray) is the fourth *baray* in the Angkor region, it measures 3,600 metres by 930 metres and has a storage capacity of 5 million cubic metres for the first phase, and for the second phase the storage can be increased to 10 million cubic metres by raising the dykes. Built in the 12<sup>th</sup> century (1181) by King Javaraman VII, it started drying up in the 16<sup>th</sup> century. It was a new invention in the technology of water engineering in the Khmer Empire; the North Baray was filled with water by a network of dykes and canals to collect run-off water and raise the water level so it flowed into the Baray. The dyke and canal system begins at the north-west dyke of the

North Baray and turns four times at 90 degrees before running north in a straight line to the foot of the Kulen mountain chain. These large canals are now used today by local people as rice fields because no water has flowed through them for more than 500 years, and some villagers have even built their houses in what is actually the disused run-off water collection system for the North Baray.

This feeding system for the North Baray is very complicated, and before our research in 2004-2005 it was very difficult to understand how this Baray was filled. Local people told us that the dyke found on the site to the north of North Baray is called the 'Ancient royal road to the mountain', but from experience we decided it could not be a royal road because:

1. It makes 90 degree turns – that would not be the case with a 12<sup>th</sup> century road.
2. It has only one canal along the dyke – in ancient times, and even today in the plains of Cambodia, the roads were built with excavated soil and the channel that was left later became a drainage canal. But these dykes have only one large canal and the other side is at natural ground level.

These reasons led us to the conclusion that they are actually the dyke and canal system that collected run-off water for the North Baray.



**Plates 3a and b**

Detail of the 9<sup>th</sup> century Bampenh Reach spillway which directed water to the Capital, Roluos (later to Angkor) showing details of the grooves in the laterite blocks. Photos: Author.

All the *barays* in the Angkor region have their water inlet at their north-eastern corner, but during the renovation of the North Baray it was necessary to move the inlet to the connection point between the feeder system and the north dyke. This decision reduces the risk of flooding for the local people who now live in the old system of water collection for the Baray. Also, the ancient canal has only been dredged for 20 metres instead of 60 metres to safeguard the local people's rice fields.

The restoration of the hydraulic system to functionality was finally started with the approval of ICC-Angkor (the International Co-ordinating Committee for Angkor) in June 2007. The North Baray collected 700,000 cubic metres of water in the first rainy season (in 2008), 2,980,000 cubic metres in 2009, 3,678,000 cubic metres in 2010, and 5,000,000 cubic metres in 2011, 2012 and 2013. The local community come to catch fish in this *baray* and can catch at least 3 kilos of fish per day per fisherman. (Plates 4 and 5) APSARA is also helping the villagers of Phlong and Leang Dai to set up a community business on the natural circuit at North Baray (the Baray Reach Dak Community Tour), and this will give them an income from guiding tourists around the natural circuit of the Baray and taking them in local boats to visit the flooded forest inside the Baray. (Plate 6)



**Plate 6**  
Tourists boating on the North Baray. Photo: Author.

### 3.3. Neak Poan

This is an island temple (376 metres by 320 metres) in the middle of the North Baray, like the Mébon (temples) of the other *barays*, but in ancient times it also functioned as a hospital using medicinal plants. The temple has five basins which, in recent times, were dry all year round except for the central basin which had water in it for only a few months during the rainy season. (Plates 7 and 8) However, since the North Baray has been refilled, these five basins have again filled with water and are full all year round. The water from the North Baray infiltrates the big central basin, then when the water reaches the



**Plates 4-5**  
Fishing in the North Baray. Photos: Author







**Plates 7-8**  
Neak Poan and its water basins. Photos: Author.



**Plate 9**  
Ceremony at Neak Poan – praying for rain, day 1 of the ceremony. Photo: Author.



**Plate 10**  
Ceremony at Neak Poan - praying for rain, day 2 of the ceremony. Photo: Author.

spillway level located in the small chapel between the central basin and the smaller basins, it starts to overflow and fills the smaller basins.

This movement of water into the five basins linked to the North Baray provides one of the best illustrations of the hydraulic system in the Angkor World Heritage property, showing that the ancient Khmer used the techniques of infiltration and exfiltration (underground flow) to recharge the groundwater, moat and basins.

The local people still come to collect medicinal plants from the island of Neak Poan, and also continue to perform some ceremonies such as asking the sacred water to cure diseases, or to give blessings to bring luck, or asking the gods for rain so the rice harvest will be

good. In Cambodia during the rainy season (May-October) there is a short dry season between mid-July and mid-August; if this break in the rain lasts for a long period that creates problems for the rice growers, so each year the local people in Phlong and Leang Dai hold a ceremony at which they ask the gods for the rain to come back. The Phlong and Leang Dai villagers believe Neak Poan is the key place for rain. The ceremony is held over two days, on the first day they invite the monks to bless them before the start of the ceremony, they offer rice to the monks and perform *trott* dancing around the five basins of the Neak Poan temple. On the second day, the monks recite *Chhyanto* to bless the villagers at the central temple and they have a lunch at the site. As shown in the picture, the rain comes suddenly at the end of the second day. (Plates 9 and 10)

#### 4. Optimisation of water management in the Angkor region

The Angkor Park is sandwiched between the Kulen mountain range and the Tonle Sap Lake. The city could have been flooded at any time during the rainy seasons or during major storms. No written sources from the time of the Khmer Empire mention either floods or droughts in the Angkor region, and nor do the Khmer people have any memory or ancient legends relating to such disasters. It would seem, therefore, that those problems did not occur in the past, indicating that the water management system in ancient times was capable of optimising water resources.

But because the ancient water system had not functioned for such a long time, and because of the deforestation on the top of Mount Kulen, in 2004 Angkor experienced a drought; the Angkor Wat moats and the Srah Srang dried up, and this was followed by floods every year from 2009 until 2013.

An understanding of the overall organisation of the Angkor hydraulic system - rivers, *barays*, moats, canals, ponds and dykes - can be found through field work, identifying how the water flows and pinpointing channels and their connections. Such field research has revealed that in the 9<sup>th</sup> century they built a laterite spillway called Bampenh Reach, and in the 12<sup>th</sup> century they constructed a laterite bridge with several arches on the Siem Reap River, with dykes running east-west to the north-east of the Ta Som temple, which could be used to control the water flow and discharge. This system also linked into the North Baray system. Utilising this distribution node, and the articulating channels, we managed to distribute water to the west (North Baray, West Baray and the Pourk River), to the East (Roluos River) and to the South (the moats of Angkor Thom and Angkor Wat and the Siem Reap River).

In 2012, the main part of this system was renovated by cleaning out the ancient canal and 17 kilometres of dykes, enabling Angkor and Siem Reap City to avoid flooding during the rainy seasons of 2012 and 2013. Without this work, Siem Reap City would have faced at least four successive waves of flooding each year. This result confirms that the ancient Angkor hydraulic system can not only optimise the management of water resources, it can also prevent flooding.

#### 5. Conclusion

Understanding of the functioning of the ancient hydraulic system was only possible with the help of local people who had inter-generational knowledge transmitted to them from previous generations. The names of the places and the words they use were very significant in assisting our research.

The rehabilitation of the ancient hydraulic systems is the best way to conserve this heritage and allows this system to function again to manage present day challenges from droughts and floods. The outcomes in 2012 and 2013 undoubtedly confirmed that these systems still work effectively today to protect Angkor from natural disasters, and from the problems caused by the increasing use of water by visitors in the region.

In the era of Angkor's prosperity, this imperial capital had an efficient, coherent system of hydraulic engineering. This work is a model of conservation and restoration, and the most efficient, cost-effective and sustainable contribution to the development of the Angkor/Siem Reap region and to the reduction of poverty in the area. 🏘️



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